

Morphic Studio

Document 19 — Storyboard Studio (Expanded Specification)

Document ID: MS-019

Version: 2.0

Status: Expanded Specification

Priority: Highest (Production Pipeline)

Purpose

The Storyboard Studio transforms written stories into structured visual production plans.

It acts as the bridge between storytelling and content generation by converting scripts into scenes, panels, camera directions, character placements, timing, dialogue, and production tasks.

Instead of immediately generating comics or animations, Morphic Studio first creates a storyboard that the creator can review, edit, and approve.

The storyboard becomes the production blueprint for every downstream module.

Vision

The Storyboard Studio should feel like working with a professional storyboard artist and film director.

The AI proposes visual interpretations, but the creator remains the director.

Every scene should be understandable before any expensive AI generation begins.

Planning first reduces cost, improves consistency, and gives creators confidence.

Core Philosophy

Plan Before Production

Instead of:

Story



Generate Comic

The workflow becomes:

Story



Storyboard



Creator Review



Approval



Production

The storyboard is the final checkpoint before large-scale AI generation.

Objectives

The Storyboard Studio shall:

- Analyze the script.
- Break stories into scenes.
- Divide scenes into shots or panels.
- Recommend camera angles.

- Recommend compositions.
 - Recommend character placement.
 - Recommend lighting.
 - Recommend emotions.
 - Estimate production complexity.
 - Prepare the project for Comic Studio or Animation Studio.
-

Story Analysis

When a story is submitted, Morphic Core shall identify:

- Scene boundaries
- Character appearances
- Dialogue
- Actions
- Camera opportunities
- Environmental changes
- Emotional beats
- Important props
- Sound cues
- Lighting conditions

The creator may adjust these suggestions before production.

Storyboard Structure

Each storyboard contains:

Project



Chapter



Scene



Shot / Panel



Dialogue



Camera



Characters



Props



Background



Notes



AI Tasks

This structure supports both comics and animation.

Storyboard Elements

Each panel or shot should include:

- Scene ID
- Panel ID
- Description
- Dialogue
- Characters Present
- Active Outfits
- Facial Expressions
- Character Poses
- Camera Angle

- Camera Distance
 - Camera Movement (Animation)
 - Lighting
 - Time of Day
 - Weather
 - Background
 - Props
 - Sound Notes
 - Music Notes
 - AI Generation Status
-

Camera Language

The Storyboard Studio should support cinematic framing such as:

- Wide Shot
- Extreme Wide Shot
- Medium Shot
- Close-Up
- Extreme Close-Up
- Over-the-Shoulder
- High Angle
- Low Angle
- Dutch Angle
- Point of View (POV)
- Tracking Shot
- Establishing Shot

AI may recommend shots, but creators have the final decision.

Character Direction

For each panel or shot, the storyboard specifies:

- Position
- Facing Direction
- Body Pose
- Hand Pose
- Facial Expression
- Eye Direction
- Emotion
- Movement
- Interaction with Props

- Interaction with Other Characters

This information improves visual consistency.

Environment Planning

Each storyboard entry references:

- Location
- Background Asset
- Weather
- Time
- Lighting
- Environmental Effects
- Crowd Density
- Atmosphere

The AI should reuse existing assets whenever possible.

User Stories

As a creator, I want to review the visual plan before generation.

As a creator, I want to rearrange scenes easily.

As a creator, I want to change camera angles without rewriting my script.

As a creator, I want to edit only one panel instead of regenerating an entire chapter.

Functional Requirements

The Storyboard Studio shall allow users to:

- Create storyboards.
- Auto-generate storyboards.
- Edit scenes.
- Reorder scenes.
- Split scenes.
- Merge scenes.
- Edit camera angles.
- Edit dialogue.

- Change character placement.
 - Assign assets.
 - Attach notes.
 - Approve storyboard.
 - Send approved storyboard to production.
-

AI Workflow

When a storyboard is created:

1. Read Script.
 2. Read Story Bible.
 3. Read Character Manager.
 4. Read World Builder.
 5. Search Asset Library.
 6. Build Storyboard.
 7. Detect missing assets.
 8. Recommend camera language.
 9. Present storyboard.
 10. Await creator approval.
 11. Export to Comic Studio or Animation Studio.
-

UI Components

The Storyboard Studio should include:

- Scene Navigator
 - Storyboard Canvas
 - Thumbnail Strip
 - Timeline View
 - Camera Editor
 - Character Placement Tool
 - Asset Browser
 - AI Suggestions Panel
 - Notes Panel
 - Validation Warnings
 - Approval Button
-

Database Entities

Core entities include:

- Storyboard
 - StoryboardScene
 - StoryboardPanel
 - CameraInstruction
 - CharacterPlacement
 - SceneNote
 - ProductionStatus
-

API Considerations

Potential endpoints:

- Create Storyboard
 - Retrieve Storyboard
 - Update Storyboard
 - Reorder Scenes
 - Validate Storyboard
 - Export Storyboard
 - Generate Production Tasks
-

Non-Functional Requirements

The Storyboard Studio should:

- Load quickly.
 - Handle large projects.
 - Autosave changes.
 - Support drag-and-drop editing.
 - Remain responsive during AI analysis.
 - Scale to feature-length productions.
-

Error Handling

If AI cannot confidently interpret a scene:

- Highlight uncertainty.

- Request clarification.
- Allow manual editing.
- Continue processing other scenes.

If required assets are missing:

- Recommend generation.
 - Mark production as incomplete until resolved.
-

Future Expansion

Future versions may include:

- Automatic animatics.
 - Voice-assisted storyboarding.
 - Gesture-based editing.
 - VR storyboard review.
 - Multi-user collaboration.
 - AI cinematography advisor.
 - Camera path visualization.
 - Interactive production scheduling.
-

Acceptance Criteria

The Storyboard Studio succeeds when creators can review, edit, and approve a complete visual production plan before any comic panels or animation frames are generated.

Key Principle

The storyboard is the bridge between imagination and production. It transforms written ideas into a structured visual language that both humans and AI can understand, ensuring every creative decision is intentional before production begins.

End of Document 19